Interface01 - Teste das funções sobre Progressões Geométricas

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As fórmulas para a **progressão aritmética** são:

Razão: $r = a_i - a_{i-1}$

Termo Geral: $a_i = a_1 + (i-1) \times r$

Soma n Elementos: $S = \frac{a_1 + a_n}{2} \times n$

As fórmulas para a **progressão geométrica** são:

Razão: $r = \frac{a_i}{a_{i-1}}$

Termo Geral: $a_i = a_1 \times r^{i-1}$

Soma n Elementos: $S = a_1 \times \frac{r^n - 1}{r - 1}$

$a_i = a_1 + (i-1)r$

$S_n = \frac{a_1 + a_n}{2} n$

```
clear
disp("------Progressão Aritmética-------")
```

```
------Progressão Aritmética-------
```

```
a_1 = 1
```

```
a_1 = 1
```

```
r = 1
```

```
r = 1
```

```
n = 100
```

```
n = 500
```

```
disp("------GerarPAritmética_v1---------")
```

```
------GerarPAritmética_v1---------
```

```
a = gerarPAritmetica_v1(a_1,r,n)
```

```
a = 1×500
     1     2     0     0     0     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     0     0     0     0     0     0     0     0     0     0 ···
a = 1×500
```

```
     1     2     3     4     0     0     0     0     0     0     0     0     0···
a = 1×500
     1     2     3     4     5     0     0     0     0     0     0     0     0···
a = 1×500
     1     2     3     4     5     6     0     0     0     0     0     0     0···
a = 1×500
     1     2     3     4     5     6     7     0     0     0     0     0     0···
a = 1×500
     1     2     3     4     5     6     7     8     0     0     0     0     0···
a = 1×500
     1     2     3     4     5     6     7     8     9     0     0     0     0···
a = 1×500
     1     2     3     4     5     6     7     8     9    10     0     0     0···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11     0     0···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12     0···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
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a = 1×500
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a = 1×500
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a = 1×500
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a = 1×500
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a = 1×500
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a = 1×500
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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a = 1×500
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a = 1×500
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a = 1×500
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a = 1×500
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a = 1×500
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a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
Elapsed time is 0.491257 seconds.
```

```
disp("------GerarPAritmética_v2---------")
```

```
------GerarPAritmética_v2---------
```

```
b = gerarPAritmetica_v2(a_1,r,n)
```

```
a = 1×500
     1     2     0     0     0     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     0     0     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     0     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     9     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10     0     0     0 ···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11     0     0···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12     0···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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a = 1×500
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a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13 · · ·
a = 1×500
```


```
     1     2     3     4     5     6     7     8     9    10    11    12    13 · · ·
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
b = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
Elapsed time is 0.434934 seconds.
```

```
disp("------GerarPAritmética_v3---------")
```

```
------GerarPAritmética_v3---------
```

```
c = gerarPAritmetica_v3(a_1,r,n)
```

```
a = 1×500
     1     2     0     0     0     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     0     0     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     0     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     0     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     0     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     0     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     0     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     9     0     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10     0     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11     0     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12     0 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
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     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13···
a = 1×500
```

```
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
a = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
c = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
Elapsed time is 0.431431 seconds.
```

```
disp("------GerarPAritmética_v4---------")
```

```
------GerarPAritmética_v4---------
```

```
d = gerarPAritmetica_v4(a_1,r,n)
```

```
d = 1×500
     1     2     3     4     5     6     7     8     9    10    11    12    13 ···
Elapsed time is 0.006033 seconds.
```

```
disp("------SomaPAritmética_v4---------")
S = SomaPAritmetica(a)
```